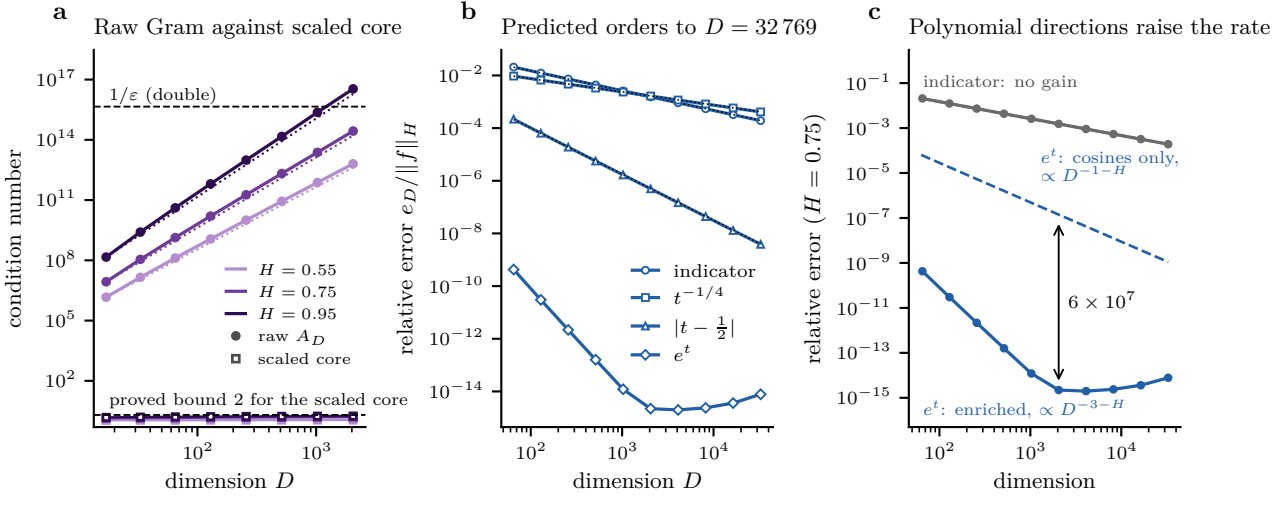

a
Raw Gram against scaled core
condition number
10^{17}
10^{14}
10^{11}
10^{8}
10^{5}
10^{2}
$1/\varepsilon$ (double)
$H = 0.55$
$H = 0.75$
$H = 0.95$
raw A_D
scaled core
proved bound 2 for the scaled core
10^{2}
10^{3}
dimension D
b
Predicted orders to $D = 32\,769$
relative error $e_D/\|f\|_H$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
indicator
$t^{-1/4}$
$|t - \frac{1}{2}|$
e^t
10^{2}
10^{3}
10^{4}
dimension D
c
Polynomial directions raise the rate
relative error ($H = 0.75$)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
indicator: no gain
e^t: cosines only, $\propto D^{-1-H}$
6×10^7
e^t: enriched, $\propto D^{-3-H}$
10^{2}
10^{3}
10^{4}
dimension